

Introduction

In this computer lab, you will use Monte Carlo particle simulations made with the Geant 4 package ¹. You will explore the structure of the data, and perform some statistical calculations on them.

1 Programming preliminaries

1.1 Two-dimensional histograms

Two-dimensional histograms are a good tool to explore correlations or distributions of your data. You can make one with numpy like this:

```
>>> import numpy as np
>>> import matplotlib.pyplot as plt
>>> xs = np.array([1,5,3,5,2])
>>> ys = np.array([0,4,2,3,1])
>>> #NOTE THE ORDER OF X AND Y:
>>> H,xbin,ybin =np.histogram2d(ys,xs,bins=5)
>>> xmin = xbin[0]
>>> xmax = xbin[-1]
>>> ymin = ybin[0]
>>> ymax = ybin[-1]
>>> im = plt.imshow(H,extent=[xmin,xmax,ymin,ymax],origin='low',cmap=plt.cm.afmhot)
>>> plt.show()
```

1.2 Finding percentiles of a numpy array

It is convenient to be able to quickly determine percentiles of a numpy array:

```
>>>> import numpy as np
>>>> x = np.array([0,5,3,2,5])
>>>> #10percentile:
>>>> print( np.percentile(x,10))
0.8
>>>> #75percentile:
>>>> print( np.percentile(x,75))
5.0
```

1.3 concatenating numpy arrays

The example below shows how you can concatenate two numpy arrays:

```
>>> import numpy as np
>>> a = np.array([[1,2],[3,4]])
>>> b = np.array([[5,6],[7,8]])
>>> c = np.concatenate([a,b])
>>> print( c)
```

¹<http://geant4.web.cern.ch/geant4/>

```
[[1 2]
 [3 4]
 [5 6]
 [7 8]]
```

1.4 Finding all files matching a pattern

Using `glob`, you can find all files matching a pattern:

```
>>> import glob
>>> glob.glob('*.txt')
['emailFK706116.txt', 'Exercises.txt', 'labAttendance16.txt']
```

1.5 Interpolating a function

You can interpolate from a function using `scipy.interpolate.interp1d`.

```
>>> import matplotlib.pyplot as plt
>>> from scipy import interpolate
>>> x = np.arange(0, 10)
>>> y = np.exp(-x/3.0)
>>> f = interpolate.interp1d(x, y)

>>> xnew = np.arange(0, 9, 0.1)
>>> ynew = f(xnew) # use interpolation function returned by 'interp1d'
>>> plt.plot(x, y, 'o', xnew, ynew, '-')
>>> plt.show()
```

2 Physics Introduction

2.1 Particles through matter

Relativistic charged particles traversing matter lose energy in three main ways:²

1. The particle may ionize atoms, kicking out an electron. This effect dominates at low energies.
2. The particle may emit transition radiation at a material boundary
3. The particle may radiate photons through e.g. Cherenkov radiation or Bremsstrahlung. This is the largest effect at the highest energies, and is inversely proportional to the particle's mass.

At high energies ($E > 5$ MeV), the dominant effect for photons is the creation of an e^+e^- pair. As particles create photons through bremsstrahlung, a high-energy electron or photon will create an e^+e^- pair, resulting in more relativistic electrons radiating photons that pair-produce etc. Electrons or photons will therefore both create a large "Electromagnetic Shower" at high enough energies.

²For an overview of particle detection, *Detectors for Particle radiation* by Konrad Kleinknecht was the lab tutors choice this time.

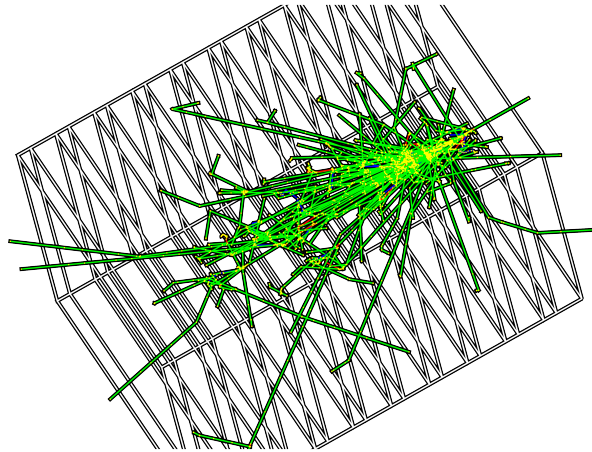


Figure 1: Electromagnetic shower in calorimeter induced by a photon

In contrast, muons, which are 200 times heavier than electrons, radiate less, and will interact much less with matter. Particle physics detectors typically identify muons this way; any charged particle that passes through the entire detector without losing much energy is classified as a muon.

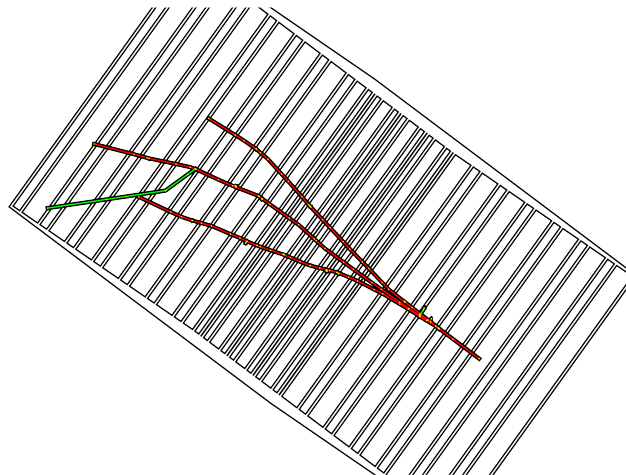


Figure 2: Muon creating a couple of secondary electrons in the calorimeter. The particles bend because of the magnetic field.

2.2 Reading the data

The data you are given is saved in numpy's `.npy` format. Each file contains an array, with five numbers for each event, and can be loaded like this:

```
>>>> import numpy as np
```

```
>>>> a = np.load("numpyfile.npy")
>>>> a.shape
(5030, 5)
>>>> print( a[0])
[125.48915443   10.00555455  109.60118788   51.44653892  100.          ]
```

In order, the numbers are:

0. E_{abs} ; The energy (in MeV) deposited in the absorbing lead plates.
1. E_{gap} ; The energy (in MeV) deposited in the argon gas.
2. L_{abs} ; The total length (in mm) of all particle tracks in the absorbing lead plates.
3. L_{gap} ; The total length (in mm) of all particle tracks in the argon gas.
4. E_{true} ; The true initial energy (in MeV) of the incoming particle.

Of these numbers, we typically measure only the energy and total track length in the argon gas³. We wish to reconstruct the initial energy of the incoming particle, or alternatively, we wish to use E_{gap} as an estimator of the total deposited energy, $E_{dep} = E_{abs} + E_{gap}$. For simplicity, you may assume in this lab that we can measure E_{abs} and L_{abs} with negligible measurement error. Three sets of files are provided where the original particle is an electron, a muon or a photon, respectively, with energies between 1000 MeV and 2000 MeV. The file names reflect the contents of the files. For instance, `RunNtup-e-1000-Etru-1000.00.npy` contains MC electron events at 1000 MeV.

2.3 Calorimeters

In this exercise, you will be given files for a lead-argon calorimeter. This technique is widely used, for example in the ATLAS detector. 10mm plates of lead absorb the shower, while the calorimeter read out in 5mm gaps between the plates. Charged shower particles traversing the liquid argon filled gap ionize the liquid, which can be read out. The energy deposited in the liquid argon is roughly equal to the sum of the length of all the particle tracks in the gap.

- Pick a subset of the simulated events from the file labelled `RunNtup-HighStatistics*.npy` and plot a scatter plot of the total deposited energy against the total track length for the electrons, the photons, and the mystery sample. What do you observe?

2.4 Monte Carlo simulation

A particle moving through matter has a probability (depending on density, material, particle type etc.) to interact in different ways. Geant4 has physics lists - a table of every interaction that is implemented. Below, you see a couple as printed in the output file:

```
eIoni:   for e-      SubType= 2
        dE/dx and range tables from 100 eV to 10 TeV in 77 bins
        Lambda tables from threshold to 10 TeV, 7 bins per decade, spline: 1
        finalRange(mm)= 1, dRoverRange= 0.2, integral: 1, fluct: 1, linLossLimit=
          0.01
        ===== EM models for the G4Region DefaultRegionForTheWorld =====
        MollerBhabha : Emin=          0 eV      Emax=          10 TeV
```

³These calorimeters are therefore called *sampling* calorimeters

```

eBrem:  for e-      SubType= 3
        dE/dx and range tables from 100 eV  to 10 TeV in 77 bins
        Lambda tables from threshold to 10 TeV, 7 bins per decade, spline: 1
        LPM flag: 1 for E > 1 GeV,  HighEnergyThreshold(GeV)= 10000
        ===== EM models for the G4Region  DefaultRegionForTheWorld =====
                eBremSB :  Emin=          0 eV    Emax=          1 GeV    DipBustGen
                eBremLPM :  Emin=          1 GeV   Emax=         10 TeV    DipBustGen

```

The sheer number of interactions, the multiple detector geometries, etc. mean that we must turn to Monte Carlo methods to compute e.g. the distributions of positrons that transverse some complex piece of matter⁴.

- Can you think of some statistical distributions needed to simulate elementary particles?

3 Lab task

3.1 Showers vs. MIPs

- Open up some of the files, and plot the histograms of the deposited energy in lead and in the gas gap for the various particles. How are they different? If you had to tell the difference between muons and electrons, could you do it?
- If yes, imagine you have found $N = 15$ particles you are sure to be muons after running your experiment an hour. What are your upper and lower limits on the expected number of muons per hour at a 90% confidence level?

3.2 Estimating electron fraction

Look at the files with `HighStatistics` in their filenames. They contain 10^5 events of photons and electrons with 100 MeV and 200 MeV incident energy respectively. Suppose you wish to find out how many photons there are compared to electrons.

- With this high-statistic data set, how would you estimate the probability density function for L_{gap} for each of the two species of particles?
- What is the expression of the likelihood function for this data set? Hint: You would need to include the number of electrons, photons, PDF for L_{gap} for electrons, photons, and remember that you do not always observe the exact number of events you expect from your model.
- Construct the likelihood function for the number of photons and electrons using the mystery MC events in `mystery_Etru_-1.00.npy`, and compute the maximum likelihood estimate of the number of photons and electrons in the mystery sample.

⁴say your head in a PET scanner, or, if you are unlucky, in a synchrotron